

Memo randu
EBN122 SEMESTER TOETS/TEST 2 (07-10-2013)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentennummer: Student number:						
Voorletters / Initials:		Taal / Language:	Afr		Eng			

Q1		Q2		Q3		Tot:	/45
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VRAAG 1 – QUESTION 1 (10)		
1.1	$i_1 = 1A \checkmark$	$i_2 = -1A \checkmark$
1.2	$i_1 = -2A \checkmark$	$i_2 = -0,333 A \checkmark$
1.3	$\checkmark \begin{bmatrix} 25 & -5 \\ -5 & 25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix} \checkmark$	
1.4	a) $I_B = 0,125mA \checkmark$	b) $V_0 = 2,4 V \checkmark$

VRAAG 2 – QUESTION 2 (20)			
2.1	$v_{0(10V)} = 6V \checkmark \checkmark$	$v_{0(1A)} = 1.2V \checkmark \checkmark$	
2.2	$V_1 = 12V \checkmark$	$R_1 = 6\Omega \checkmark$	$V_2 = 12.5V \checkmark$ $R_2 = 2.5\Omega \checkmark$
2.3	$V_{Th} = 8V \checkmark \checkmark$	$R_{Th} = 0.67\Omega \checkmark \checkmark$	
2.4	$I_N = 5A \checkmark \checkmark$	$R_N = 8\Omega \checkmark \checkmark$	
2.5	$P = 42.6 W \checkmark \checkmark$	$R = 4\Omega \checkmark \checkmark$	

VRAAG 3 – QUESTION 3 (15)			
3.1	$a) V_0 = 2,4V \checkmark \checkmark$	$b) V_0 = -5V \checkmark \checkmark$	
3.2	$a) V_A = -9V \checkmark \checkmark$	$b) R_F = 20k\Omega \checkmark \checkmark$	
3.3	$a) V_X = -2V \checkmark \checkmark$	$b) V_Z = 6V \checkmark \checkmark$	
3.4	$V_1 = 2,5V \checkmark$	$V_0 = -13,8V \checkmark$	$I_0 = 1,8125mA \checkmark$